

ORION

PART 09

Cloud, Edge & Infrastructure

Data centers, Kubernetes, edge nodes, distributed deployment

Technical Architecture Series • Beginner → Professional

How to use this volume

This document is one independent volume in the twelve-part ORION technical series. Beginners should read from the start; experienced readers can jump directly to the relevant architecture sections.

ORION is a secure communications, infrastructure-coordination, emergency-response, and decision-support architecture. High-impact weapons-control functionality is deliberately outside this design.

Infrastructure strategy

Use centralized regional services for coordination and edge nodes for local resilience and latency-sensitive functions.

Professional lens: define ownership, interfaces, trust boundaries, failure behavior, security controls, observability, and measurable success criteria before implementation.

Cloud building blocks

Compute, object storage, databases, messaging, load balancers, identity, monitoring, and secrets management.

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Containers

Package applications with their dependencies for repeatable deployment across environments.

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Kubernetes

A container orchestration platform that manages scheduling, service discovery, scaling, health checks, and rolling updates; use it when its benefits justify the operational complexity.

Professional lens: define ownership, interfaces, trust boundaries, failure behavior, security controls, observability, and measurable success criteria before implementation.

Cloud-edge model

Central/regional cloud → control plane and shared services → regional services → edge gateway → local cache/services/inference → field devices.

Professional lens: define ownership, interfaces, trust boundaries, failure behavior, security controls, observability, and measurable success criteria before implementation.

Deployment

Infrastructure as code, automated testing, progressive deployment, health checks, rollback, and configuration versioning.

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Multi-region

Use independent failure domains. Decide explicitly which data may replicate asynchronously and which operations need stronger coordination.

Professional lens: define ownership, interfaces, trust boundaries, failure behavior, security controls, observability, and measurable success criteria before implementation.

Cost optimization

Right-size compute, autoscale, archive cold data, control telemetry volume, choose managed services selectively, and measure cost per transaction/user.

Professional lens: define ownership, interfaces, trust boundaries, failure behavior, security controls, observability, and measurable success criteria before implementation.

Architecture Review Checklist

- Requirements and assumptions are explicit.
- Component ownership and boundaries are documented.
- Normal, degraded, and failure states are defined.
- Security and privacy are architectural requirements.
- Capacity and scaling assumptions are measurable.
- Logs, metrics, traces, and audit events cover critical flows.
- Dependencies have explicit contracts and failure behavior.
- Important decisions record alternatives and trade-offs.
- Testing covers expected behavior and controlled failures.
- Operational runbooks exist for important incidents.

Key Takeaways

Cloud, Edge & Infrastructure should be implemented as a bounded layer of ORION, with explicit interfaces to the rest of the platform.

The goal is not maximum complexity. The goal is measurable resilience, security, maintainability, interoperability, and operational usefulness.

Glossary

- API — Application Programming Interface used for software-to-software communication.
- Edge — Computing placed near users or physical systems.
- Event — A record that something happened.
- Failover — Switching to a redundant component after failure.
- Idempotency — Repeating an operation does not create an unintended additional effect.
- NTN — Non-terrestrial network, commonly involving satellite infrastructure.
- Policy — A rule that determines what is allowed under stated conditions.
- RPO — Recovery Point Objective; acceptable data-loss window.
- RTO — Recovery Time Objective; target recovery time.
- Zero trust — Continuous verification instead of implicit trust based on network location.